Mathematics of Machine Learning

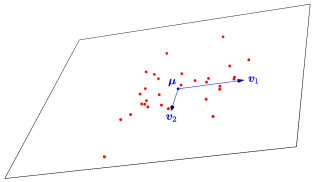
Matrix Factorisation

Name	Titanic	Dr No	The Matrix	Spartacus	Ten
Alice	2	Ø	4	5	2
Bob	5	3	Ø	3	Ø
Chas	Ø	2	3	5	3

Matrix completion, low-rank approximation, recommender systems, alternating least squares, non-negative matrix factorisation

Outline

1. **Missing Data and Recommenders**
2. Low-Rank Approximation
3. Probabilistic Approach
4. Non-Negative Matrix Factorisation



Matrix Completion

- It is extremely common to be given a matrix of data with missing entries (particularly when the data comes from humans filling in forms)
- Most learning machines need a complete set of data, so we either throw away examples or features with missing values (sometimes leaving nothing) or do our best to infer them
- Inferring them only works if there is structure in the matrix that we can exploit
- The structure we use in this lecture: **structured data typically lies close to a low-dimensional subspace** (the idea behind PCA and the SVD)

Recommender Systems

- The best-known example: recommending items to users, which can be financially important
  - ⋆ Recommendations are reported to drive a large share of Amazon's sales
  - ⋆ Netflix (then a DVD rental company, now a streaming service) ran a $1 million competition (2006–09) to improve the quality of its recommendations
- **Content-based filtering** uses (often text) features to find similar items; **collaborative filtering** uses ratings supplied (implicitly or explicitly) by users
- We concentrate on collaborative filtering

Recommending Movies

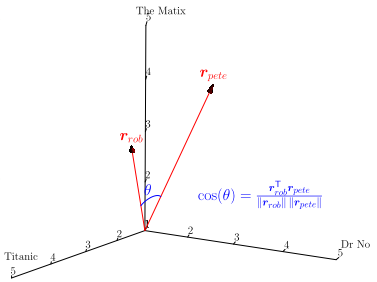
- The classic example is to recommend a movie given some ratings from users

Name	Titanic	Dr No	The Matrix	Spartacus	Ten
Alice	2	Ø	4	5	2
Bob	5	3	Ø	3	Ø
Chas	Ø	2	3	5	3

- Ø means there are no ratings
- The task is to predict the unknown ratings and make a recommendation
- Typically data sets are huge and very sparse

Classic Collaborative Filtering

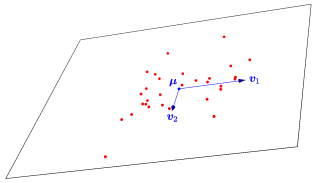
- The classic collaborative filtering technique is to find similar users (those giving similar ratings)
- Use the recommendation of similar users weighted by their similarity
- Can also recommend items that are similar (i.e. items that have received similar ratings) to those that a user rates highly



$$\hat{R}_{ui} = \frac{\sum_{u'} \text{sim}(u,u') R_{u'i}}{\sum_{u'} \text{sim}(u,u')}$$

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Matrix Completion

- A different (often better) method is to think of recommendation as a matrix completion task

Name	Titanic	Dr No	The Matrix	Spartacus	Ten
Alice	2	Ø	4	5	2
Bob	5	3	Ø	3	Ø
Chas	Ø	2	3	5	3

- Complete the matrix

$$\mathbf{R} = \begin{pmatrix} 2 & & 4 & 5 & 2 \\ 5 & 3 & & 3 & \\ & 2 & 3 & 5 & 3 \end{pmatrix}$$

Replace by Means

- The simplest guess uses the means, column-wise and row-wise

$$R_{ui} \approx \bar{R}_u + \bar{R}_i - \bar{R}$$

where

  - $\bar{R}_u$ is the mean rating of user u
  - $\bar{R}_i$ is the mean rating for item i
  - $\bar{R}$ is the mean overall rating
- Gets the means right and gives a reasonable rating
- (For general missing data: replace by the feature's mean)
- But this ignores structure: similar users give similar ratings

Best Low-Rank Approximation

- Recall the SVD of a complete matrix, with $s_1 \geq s_2 \geq \cdots \geq 0$

$$\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\mathsf{T} = \sum_k s_k \boldsymbol{u}_k \boldsymbol{v}_k^\mathsf{T}$$

- Keeping the K largest singular values gives a rank-K matrix $\mathbf{X}_K = \sum_{k=1}^K s_k \boldsymbol{u}_k \boldsymbol{v}_k^\mathsf{T}$
- **Eckart–Young theorem**: $\mathbf{X}_K$ is the best rank-K approximation to $\mathbf{X}$, with error $\|\mathbf{X} - \mathbf{X}_K\|_F^2 = \sum_{k>K} s_k^2$ (also best in the spectral norm, with error s_{K+1})
- Data close to a K-dimensional subspace has small discarded s_k (this is effectively PCA)

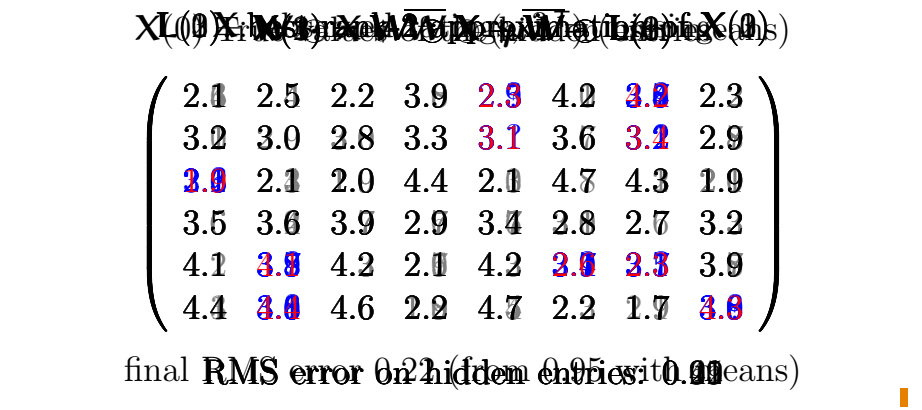
Completing a Matrix with the SVD

- The SVD needs a complete matrix, so we iterate
  1. Fill in the missing entries (e.g. with the user means)
  2. Compute the best rank-K approximation $\mathbf{L}$ of the filled-in matrix
  3. Replace the missing entries (only) by the values in $\mathbf{L}$
  4. Repeat from 2
- With a mask $\boldsymbol{W}$ ($W_{ij} = 1$ where X_{ij} is observed, 0 otherwise) and $\odot$ the element-by-element product

$$\mathbf{X}(t+1) = \boldsymbol{W} \odot \mathbf{X} + \overline{\boldsymbol{W}} \odot \mathbf{L}(t)$$

where $\overline{\boldsymbol{W}} = 1 - \boldsymbol{W}$

SVD in Operation



Low-Rank Approximation

- More directly, approximate an $m \times n$ matrix $\mathbf{X}$ by

$$\mathbf{X} \approx \mathbf{A}\,\mathbf{B}$$

  where $\mathbf{A}$ is $m \times K$ and $\mathbf{B}$ is $K \times n$
- $\mathbf{AB}$ has *rank* at most K
- We replace $m \times n$ elements by $(m+n) \times K$ free parameters
- K controls the complexity of the model: large K fits the data better but may over-fit
- Choose K by cross-validation (hide some of the observed entries and see how well we predict them)

Low-Rank Updates: LoRA

- **LoRA** (low-rank adaptation, Hu et al. 2021) uses the same idea to fine-tune a large network: a pretrained weight matrix $\boldsymbol{W}_0$ $(m \times n)$ becomes $\boldsymbol{W}_0 + \Delta\boldsymbol{W}$
- LoRA freezes $\boldsymbol{W}_0$ and learns only a low-rank change

$$\Delta\boldsymbol{W} = \mathbf{AB}, \qquad \mathbf{A}\ (m \times K),\ \mathbf{B}\ (K \times n),\ K \ll m,n$$

  with one factor starting at zero (so training starts from $\boldsymbol{W}_0$)
- For $m = n = 4096$ and $K = 8$: $(m+n)K \approx 6.6 \times 10^4$ parameters to train instead of $mn \approx 1.7 \times 10^7$ (0.4%)
- The bet: the change a new task needs is close to low rank; each task stores only its own $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{AB}$ can be added into $\boldsymbol{W}_0$ afterwards

Least-Squares

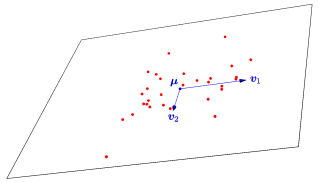
- Assume we have elements of $\mathbf{X}$ only at the locations $\mathcal{D}$
- We can seek a least-squares solution for $\mathbf{A}$ and $\mathbf{B}$

$$(\mathbf{A},\mathbf{B}) = \underset{(\mathbf{A},\mathbf{B})}{\operatorname{argmin}} \sum_{(i,j)\in\mathcal{D}} \left(X_{ij} - \sum_{k=1}^{K} A_{ik} B_{kj} \right)^2$$

- OK, but it can over-fit
- $\mathbf{A}$ and $\mathbf{B}$ are never unique: $\mathbf{AB} = (\mathbf{AG})(\mathbf{G}^{-1}\mathbf{B})$ for any invertible $K \times K$ matrix $\mathbf{G}$

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Bayes' Rule and MAP

- For parameters $\boldsymbol{\theta}$ and data $\mathcal{D}$, Bayes' rule gives the **posterior** (likelihood $\times$ prior / evidence)

$$\mathbb{P}(\boldsymbol{\theta}|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\boldsymbol{\theta})\mathbb{P}(\boldsymbol{\theta})}{\mathbb{P}(\mathcal{D})}$$

- The **maximum a posteriori** (MAP) solution is the most probable $\boldsymbol{\theta}$ ($\mathbb{P}(\mathcal{D})$ does not depend on it)

$$\boldsymbol{\theta}_{\text{MAP}} = \operatorname*{argmax}_{\boldsymbol{\theta}} \Big(\log(\mathbb{P}(\mathcal{D}|\boldsymbol{\theta})) + \log(\mathbb{P}(\boldsymbol{\theta})) \Big)$$

- With a flat prior it is the maximum-likelihood solution
- Limitations: MAP gives a single point, throwing away our uncertainty about $\boldsymbol{\theta}$; and the peak of a density depends on how we parametrise $\boldsymbol{\theta}$

Bayesian Approach

- We can use a Bayesian approach to infer $\mathbf{A}$ and $\mathbf{B}$
- The likelihood is

$$\mathbb{P}(X_{ij}|\mathbf{A},\mathbf{B}) = \mathcal{N}\left(X_{ij} \middle| \sum_{k=1}^{K} A_{ik}B_{kj}, \sigma^2 \right)$$

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}$$

- σ measures the natural error in the ratings

Priors

- We also need priors

$$\mathbb{P}(\mathbf{A}) = \prod_{i,k} \mathcal{N}(A_{ik}|0,\sigma_A^2), \qquad \mathbb{P}(\mathbf{B}) = \prod_{k,j} \mathcal{N}(B_{kj}|0,\sigma_B^2)$$

- σ_A^2 and σ_B^2 explain the variation in the original matrix $\mathbf{X}$
- Note that

$$\mathbb{E}\left[\sum_{k=1}^{K} A_{ik}B_{kj} \right] = 0, \qquad \mathbb{V}\mathrm{ar}\left(\sum_{k=1}^{K} A_{ik}B_{kj} \right) = K\sigma_A^2\sigma_B^2$$

  which should be equal to the variance in X_{ij}

MAP Solution

- Calculating the posterior probability for $\mathbf{A}$ and $\mathbf{B}$ is intractable
- We can, however, compute the *maximum a posteriori* solution

$$\begin{aligned}\mathcal{E} &= \sum_{(i,j)\in\mathcal{D}} \log(\mathbb{P}(X_{ij}|\mathbf{A},\mathbf{B})) + \log(\mathbb{P}(\mathbf{A})) + \log(\mathbb{P}(\mathbf{B})) + C \\ &= -\frac{1}{2\sigma^2} \sum_{(i,j)\in\mathcal{D}} \left(X_{ij} - \sum_{k=1}^{K} A_{ik}B_{kj} \right)^2 - \frac{1}{2\sigma_A^2} \sum_{i,k} A_{ik}^2 \\ &\quad - \frac{1}{2\sigma_B^2} \sum_{k,j} B_{kj}^2 + C'\end{aligned}$$

Iterative Learning

- We can maximise the log-posterior by first maximising the log-posterior with respect to $\mathbf{A}$ keeping $\mathbf{B}$ fixed

$$\frac{\partial \mathcal{E}}{\partial A_{ik}} = \frac{1}{\sigma^2} \sum_{j|(i,j)\in\mathcal{D}} \left(X_{ij} - \sum_{l=1}^{K} A_{il} B_{lj} \right) B_{kj} - \frac{1}{\sigma_A^2} A_{ik} = 0$$

- This is equivalent to

$$\sum_{l=1}^{K} \left(\frac{\sigma^2}{\sigma_A^2} \delta_{k,l} + \sum_{j|(i,j)\in\mathcal{D}} B_{lj} B_{kj} \right) A_{il} = \sum_{j|(i,j)\in\mathcal{D}} B_{kj} X_{ij}$$

Solving for $\mathbf{A}$

- We got

$$\sum_{l=1}^{K} \left(\frac{\sigma^2}{\sigma_A^2} \delta_{k,l} + \sum_{j|(i,j)\in\mathcal{D}} B_{lj} B_{kj} \right) A_{il} = \sum_{j|(i,j)\in\mathcal{D}} B_{kj} X_{ij}$$

- Which has the form $\mathbf{M}^i \boldsymbol{a}^i = \boldsymbol{c}^i$

$$M^i_{kl} = \frac{\sigma^2}{\sigma_A^2} \delta_{k,l} + \sum_{j|(i,j)\in\mathcal{D}} B_{lj} B_{kj}, \qquad a^i_l = A_{il}, \qquad c^i_k = \sum_{j|(i,j)\in\mathcal{D}} B_{kj} X_{ij}$$

- Solution is $\boldsymbol{a}^i = (\mathbf{M}^i)^{-1} \boldsymbol{c}^i$ (one $K \times K$ solve for each row i)

Each Step is Ridge Regression

- Write $\boldsymbol{b}_j$ for column j of $\mathbf{B}$ and $\boldsymbol{a}_i$ for row i of $\mathbf{A}$ (as a column vector)
- With $\mathbf{B}$ fixed, row i of $\mathbf{A}$ minimises

$$\sum_{j|(i,j)\in\mathcal{D}} \left(X_{ij} - \boldsymbol{a}_i^\top \boldsymbol{b}_j \right)^2 + \eta \|\boldsymbol{a}_i\|^2, \qquad \eta = \frac{\sigma^2}{\sigma_A^2}$$

- This is ridge regression: features $\boldsymbol{b}_j$, targets X_{ij}, regularisation strength η
- Its solution $\boldsymbol{a}_i = \left(\eta \mathbf{I} + \sum_j \boldsymbol{b}_j \boldsymbol{b}_j^\top \right)^{-1} \sum_j \boldsymbol{b}_j X_{ij}$ is exactly $(\mathbf{M}^i)^{-1} \boldsymbol{c}^i$
- Alternating between $\mathbf{A}$ and $\mathbf{B}$ is called **alternating least squares** (ALS): each step is convex, even though the whole problem is not

Converging to a Solution

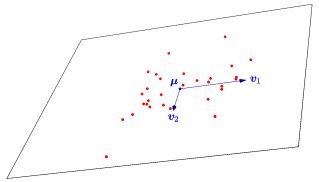
- Then solve for $\mathbf{B}$ (almost identical to solving for $\mathbf{A}$)
- Iterate between $\mathbf{A}$ and $\mathbf{B}$ until they converge
- No step can decrease the log-posterior, so it converges, usually rapidly (to a local optimum: the full problem is not convex)
- Used to predict an X_{ij} where $(i,j) \notin \mathcal{D}$
- Matrix factorisation was at the heart of the Netflix Prize winners and is still a strong baseline for recommenders

Embeddings

- Row i of $\mathbf{A}$ is a K-dimensional vector describing user i; column j of $\mathbf{B}$ describes item j
- The predicted rating is their inner product, $\boldsymbol{a}_i^\top \boldsymbol{b}_j$
- Items with similar vectors are rated alike: we have learnt an **embedding** of users and items
- Modern recommenders learn embeddings with neural networks, mostly from **implicit feedback** (clicks, views, purchases): a missing entry is then weak evidence of not liking, not simply unknown
- Matrix completion is also used to fill in missing survey, sensor and medical data

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Matrix Factorisation

- Although we have treated matrix factorisation as a matrix completion task we can also think of it as a data modelling task
- Consider a set of books in mathematics, physics and biology
- We can build a matrix of books (columns) versus words (rows), with elements being the word count
- Can use the SVD, taking $\boldsymbol{A}$ as the first three columns of $\mathbf{US}$ and $\mathbf{B}$ as the first three rows of $\mathbf{V}^\mathsf{T}$

SVD

	Euclid	Gauss	Feynman	Newton	Darwin	Mendel
theorem	51	101	5	10	3	1
proof	51	103	2	8	0	0
sum	22	51	60	72	5	7
vector	0	70	90	50	3	3
wave	0	1	51	92	2	0
species	0	4	5	1	161	89
fitness	0	1	3	1	35	2

$\approx$

	1	2	3
theorem	9.0	-2.2	-6.6
proof	8.8	-2.5	-7.0
sum	10.2	-1.7	3.4
vector	11.2	-2.3	3.4
wave	6.5	-1.3	7.7
species	5.2	17.7	-0.5
fitness	1.0	3.0	0.0

	Euclid	Gauss	Feynman	Newton	Darwin	Mendel
1	2.5	6.9	4.5	4.5	2.2	1.3
2	-0.8	-1.9	-0.9	-1.1	8.5	4.5
3	-3.5	-5.6	4.9	5.7	-0.3	-0.1

- Best rank-3 approximation, but the factors mix signs: hard to interpret

Non-Negative Matrix Factorisation

- When $\mathbf{X}$ is non-negative we can get a sparse model by minimising $E(\mathbf{A},\mathbf{B}) = \|\mathbf{X} - \mathbf{AB}\|_F^2$ with all elements of $\mathbf{A}$ and $\mathbf{B}$ non-negative
- Projected gradient descent: take a gradient step, then set negative elements to zero
- Or Lee and Seung's **multiplicative updates** ($\oslash$ is element-by-element division)

$$\mathbf{B} \leftarrow \mathbf{B} \odot (\mathbf{A}^\mathsf{T}\mathbf{X}) \oslash (\mathbf{A}^\mathsf{T}\mathbf{AB}), \quad \mathbf{A} \leftarrow \mathbf{A} \odot (\mathbf{XB}^\mathsf{T}) \oslash (\mathbf{ABB}^\mathsf{T})$$

which never make an element negative

Non-Negative Matrix Factorisation

	Euclid	Gauss	Feynman	Newton	Darwin	Mendel
theorem	51	101	5	10	3	1
proof	51	103	2	8	0	0
sum	22	51	60	72	5	7
vector	0	70	90	50	3	3
wave	0	1	51	92	2	0
species	0	4	5	1	161	89
fitness	0	1	3	1	35	2

$\approx$

	1	2	3
theorem	10.3	0.3	0.2
proof	10.5	0.1	0.0
sum	4.4	6.7	0.6
vector	5.3	7.1	0.4
wave	0.0	7.5	0.0
species	0.0	0.0	16.2
fitness	0.0	0.1	2.8

	Euclid	Gauss	Feynman	Newton	Darwin	Mendel
1	4.4	10.0	0.6	0.2	0.0	0.0
2	0.0	1.1	9.1	10.0	0.0	0.2
3	0.0	0.3	0.3	0.0	10.0	5.4

- The three factors pick out mathematics, physics and biology

Non-Negative Matrix Factorisation: Caveats

- NMF is very popular because its factors are often easier to interpret than those of the SVD
- But
  - ⋆ It cannot fit better than the SVD: by Eckart–Young the truncated SVD is the best rank-K fit (here relative errors 0.164 against 0.158)
  - ⋆ The problem is not convex: the answer depends on the starting point and on K
  - ⋆ The factors are not unique, and a "parts-based" reading of them is not guaranteed
- Always compare against simple baselines (e.g. the SVD or PCA)

Conclusion

- Structured data typically lies close to a low-dimensional subspace, and we can exploit this
- Matrix factorisation has many applications including recommender systems, data completion, and data modelling
- The truncated SVD gives the best low-rank approximation; iterating it fills in missing entries
- A Bayesian (MAP) approach helps with over-fitting, and each step of the resulting ALS is ridge regression
- Non-negative matrix factorisation gives more interpretable, though not necessarily better, factors